

EZHILAN CHINNASAMY

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Data Engineer — SQL & Data Modeling · Pipeline Automation · Multi-Agent Integration

SKILLS

- **Data Engineering:** ETL/ELT design, data modeling, pipeline orchestration, medallion architecture
- **Languages & Databases:** Python, SQL (T-SQL, PostgreSQL, MS SQL Server), PySpark, pandas, scikit-learn
- **Cloud & Platform:** Azure, GCP (Vertex AI, BigQuery, Cloud Run), AWS (EMR, S3), Docker
- **AI & Agent Integration:** LangGraph, Google ADK, Vertex AI, MCP, LangChain, litellm, RAG, RAGAS evaluation, NL2SQL
- **Payments Domain:** ACH/EFT settlement, transaction reconciliation, financial data integrity, regulatory compliance

WORK EXPERIENCE

Pay with Spire, Business Intelligence Analyst Sep 2024 – Jun 2026 | Dallas, Texas, US

- Designed a bronze/silver/gold Azure data lake fed by ADF pipelines processing 2M records daily across multiple systems — the single data surface behind multi-team wallboards, marketing automation, and AI agents.
- Scripted Azure Python runbooks replacing a human-in-the-loop reporting chain across reconciliation, billing, and risk — cutting weekly reporting errors 15% and turnaround 65%.
- Engineered a bulk merchant onboarding app collapsing 7 partner-specific workflows into one normalized ingestion UI — eliminating ~8.5 hrs per week of manual work.
- Segmented the consumer base into 12 behavioral personas, driving an A/B-tested marketing program that boosted CVR 12% and shaped C-suite acquisition strategy.
- Modeled cross-channel migration with a two-phase classification-regression pipeline, converting outputs into prioritized acquisition targeting that raised segment ARPU 30%.

DATA AND AI:

- Led delivery of an AI-powered credit/debit reconciliation system, designing the domain architecture, managing vendor API/SFTP integration and coordinating a 3-team, cross-vendor build, freeing \$160K/year of manual-labor budget.
- Built a production multi-agent analytics layer over the warehouse on Google ADK and Vertex AI — offloading ~30% of data-team workload to self-serve querying via hierarchical NL2SQL + BQML sub-agents, context engineering, and RAG retrieval.

The University of Texas at Dallas, Student Worker Jun 2024 – Sep 2024

- Developed an automated data integration pipeline harmonizing public records, then trained 50+ student researchers on it, accelerating project delivery 30%.
- Delivered a responsive timeline application mapping 5 decades of Texas tech history from 100k+ fragmented data points — one interface adopted across departments to streamline research workflows.

Ernst & Young Global Delivery Services, Software Engineer Feb 2021 – Oct 2022

- Built and scaled distributed PySpark batch pipelines on AWS EMR to process multi-terabyte unstructured log files, optimizing Hadoop cluster resource allocation to drive a 24% increase in pipeline data throughput.
- Architected a secure cloud migration framework to transition legacy, multi-tenant database infrastructure into an AWS S3 data lake, boosting query performance by 35%.

PROJECTS

Fintech Hybrid RAG Engine

- Hybrid retrieval pipeline over PostgreSQL (BM25 + pgvector HNSW, reciprocal rank fusion) for ACH compliance Q&A — +47% recall from clause-level over recursive chunking, tuned for compliance recall.
- Instrumented a RAGAS evaluation harness (faithfulness 0.829) that traced retrieval failures to BM25 index pollution and resolved them through query-type-aware weighting.

Multi-Agent Financial Reconciliation Engine

- Deterministic matcher fronting a LangGraph multi-agent investigator with MCP tools and human-in-the-loop gate; 83.3% accuracy across ~80% of volume, reserving agent reasoning for exceptions.
- Ground-truth evaluation harness catching two regressions from plausible-looking fixes; measured confidence inversely correlated with correctness, dropping self-reported confidence from routing.

EDUCATION

The University of Texas at Dallas, MS in Business Analytics and Artificial Intelligence Jan 2023 – Dec 2024 | GPA: 3.7

Certifications issued by UTD: Applied Machine Learning Certificate 

Kumaraguru College of Technology, Bachelors in Computer Science and Engineering Jul 2017 – Jun 2021 | GPA: 8.1